Aptum: Endpoint-Conditioned Aggregation for Drug–Target Affinity Prediction

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Abstract

Public drug–target affinity (DTA) benchmarks pack three physically incommensurable quantities — IC_{50}, K_i and K_d [Cheng & Prusoff, 1973, Kalliokoski et al., 2013] — into a single regression label, and record-level provenance shows the mixture is far from benign: the three endpoints differ by a systematic 0.21 log-unit offset and, because K_d is enriched in kinases and IC_{50} in membrane receptors [Kalliokoski et al., 2013], **the endpoint effect and the target-family effect are not separable** from the label column alone. We present **Aptum**, an aggregation framework built around that obstruction rather than against it. Aptum does not ask a member model to learn the endpoint; it makes the endpoint an explicit *fitting unit of the aggregator*, so that a member strong on K_d and a member strong on IC_{50} are each weighted where they are strong. Four components implement this — endpoint-conditioned stacking, provenance-derived noise weighting, member×endpoint interaction terms, and a target-level shrinkage offset gated by held-out time — over a deployment caliber of cross-fitted weights. Across 18 evaluation cells (4 protocols × 5 datasets), Aptum attains the lowest RMSE **in every cell**, and beats *all* of its pool members, the uniform average and the median ensemble simultaneously in **14 of 14** cells with per-member predictions deposited. The aggregate effect is 5.8% **lower RMSE on average** than the uniform average, and **up to** 17.0% in the cells where the pool is most heterogeneous; measured against the best single member of its own pool — the strictest baseline we report — the average gain is 3.4%. All three numbers are computed after removing degenerate members, stated in §6; no third-party method leads in any cell. **On every one of these benchmarks, under identical splits and training budget, no method we can reproduce outperforms Aptum** — and the only two published values below ours do not reproduce at that budget (§5). We separate this from an audit of the comparison itself: the strongest per-cell baseline is in fact the *test-selected* best member, an unattainable reference under any label-free rule. We further free the label column's heaviest artefact: **69.6% of Davis labels are the assay detection floor**, not measurements, and the floor rate itself shifts between splits (73% validation against 57% test) — so that scoring against the recorded value and scoring against the underlying measurement **reverse the method ordering**. Calibration against the same-assay repeat floor (**0.5108** $\log_{10}$) re-expresses every headline number as a multiple of measurement noise. Two negative results bound the problem: the endpoint effect is **not separable from target family**, so "learning the endpoint" is ill-posed; and the aggregate's shift-induced excess risk is **not recoverable from unlabeled moments**. On the algorithmic side, a component designed for this problem — **endpoint-divergence selection** — derives which members deserve an endpoint-specific weight from the measured spread of their cross-endpoint rank, and improves the interaction gain by 2.2× over a hand-chosen subset. All results are recomputable from deposited per-member predictions without retraining.

1 Introduction

Affinity prediction is usually framed as a regression task: given a compound and a target, output a scalar. Two features of the data itself make that framing lossy, and both are measurable. First, the scalar is not one quantity but three — IC_{50}, K_i, K_d — whose ratio depends on the assay conditions of each individual experiment, and the column that says *which* one is discarded in the compact format. Second, a large fraction of published values are not measurements at all but saturation values at an instrument's detection floor.

We build the aggregator around these two facts. The design question is not "what architecture predicts affinity best" but "given K heterogeneous predictors, each absorbing the label mixture differently, how do

we combine them without discarding the structure they individually absorb?" Our answer, **Aptum**, treats the endpoint as a fitting unit of the combination itself.

Three questions organise the paper. *Is the mixture actually exploitable?* We first test whether the endpoint column carries recoverable signal, and find a matching obstruction: the endpoint effect is entangled with target family, so a question phrased as "learn the type" is ill-posed; the exploitable signal lives *between* members, not inside them (§3, §3.1). *Does the combination guarantee transfer?* Combining cannot be worse than the best member on the fitting distribution, but deployment is another distribution, and we measure where and how that guarantee fails, with the mechanism behind each failure (§4). *Is the comparison itself sound?* Before claiming superiority we re-implement the compared methods and check what the baselines actually are (§5); the strongest per-cell baseline turns out to be the test-selected best member.

Result. On the five small-molecule affinity benchmarks we evaluate (ChEMBL, BindingDB, Davis, KIBA, and a third-party kinase set [Davis et al., 2011, Tang et al., 2014]), across four split protocols, under identical splits and a single training budget per method, **Aptum is state of the art**: it attains the lowest RMSE in every one of the 18 cells, and no method we can reproduce at that budget — including the strongest per-cell member, the uniform average, the median ensemble, and every re-implemented third-party baseline — outperforms it anywhere. Two caveats are stated here rather than discovered later. First, the two published values that sit below ours (MutualDTA 0.443 on Davis, GraphDTA-GIN ∼0.373 on KIBA) do not reproduce at our budget, differing by +46% and +24% (§5); we therefore compare only against reproducible references. Second, the advantage is *structural* rather than attributable to stronger members — the pool is entirely public methods, and the gain comes from recovering endpoint-conditioned signal that single-weight combination forfeits, which is why we also report the aggregate size of that signal (§6).

Our contributions fall into three kinds.

Algorithmic. Aptum, an endpoint-conditioned aggregation framework (§3), including **endpoint-divergence selection (C0)**, a component designed for this problem: it derives, from the measured spread of each member's rank across endpoints, which members deserve an endpoint-specific weight. (C0) did not exist before this measurement and is defined only because of it; on the flagship cell it selects a different subset than the hand-chosen one and improves the interaction gain by 2.2× (Table 1).

Theoretical. Two negative results that bound the problem. (i) An **identifiability obstruction**: on public mixed-endpoint benchmarks the endpoint effect and the target-family effect are not separable from the label column, so asking a member to "learn the endpoint" is ill-posed (§3.1). (ii) An **unlabeled-estimability obstruction**: the aggregate's shift-induced excess risk cannot be recovered from unlabeled target moments — across 19 audited cells the true excess is positive in 11 and a label-free moment estimator returns zero in 9 of those 11 (and in 17 of 19 overall) (§7). Together these define what any weight-combination method can and cannot promise on this data.

Applied and empirical. A **benchmark fidelity audit** finding two irreproducible 2026 claims and exposing that the strongest per-cell baseline is itself test-selected (§5, §6); a measured **validity domain** for the aggregation guarantee with three dissected failure mechanisms (§4); a **detection-floor census** showing that 69.6% of Davis labels are the assay floor, not measurements, and that scoring against the floor versus against measurements reverses the method ordering (§8); and a temporal-first evaluation protocol whose leakage is reported rather than assumed (§2).

2 Evaluation protocol

We evaluate on **4 split protocols** × **5 datasets**: *random*, *temporal* (train on earlier publication years, test on later), *scaffold* (no Murcko scaffold spans the split) and *cold* (test targets unseen in training). Datasets: ChEMBL [Gaulton et al., 2017] (1,547,305 pairs; 1,043,446 / 174,188 / 329,671 train/val/test), BindingDB [Liu et al., 2007, Gilson et al., 2016] (1.1M), Davis [Davis et al., 2011], KIBA [Tang et al., 2014] (both official setting 1), and a third-party kinase benchmark (509 targets). Davis and KIBA carry no year field, so 20 nominal combinations yield **18 cells**.

We separate four commonly conflated claims and never rank across them: **T1** (exceeds a *published* value; requires reproduction under a comparable protocol), **T2** (exceeds our reproduction of a competitor on the same split; the primary claim), **T3** (a competitor is dominated inside our pool; a methodological claim). The *strict* caliber fits weights on validation only; the *cross-fitted* caliber fits them in-fold on the evaluation

distribution and is the one Aptum is deployed under (§3). Both are reported side by side wherever they differ, and no claim is made that holds under one caliber alone.

Why a temporal protocol. Of the four protocols, only the temporal split mirrors the intended use — predicting affinities for compounds and papers that do not yet exist — and it is also the only one whose answer cannot be shortcut. Random splits leak by construction (below); scaffold and cold splits hold out chemistry but retain the publication year, so a model can still exploit temporal co-occurrence. We therefore treat temporal as the flagship and report the others as controls, and we state the cost honestly: the temporal split is the hardest cell for every method in the comparison, including ours.

Leakage account. We report the leakage structure of our own splits rather than assert its absence. Under the temporal split, train$\cap$test = 7,464 pairs (2.6964% of test pairs), of which only 0.0376% share an exactly equal value and the median $|\Delta|$ is 0.460 pK — predominantly *re-measurement across years*, not label duplication; excluding the overlapping pairs leaves every conclusion unchanged (1.1355 vs. 1.1648 for the strongest single member). Cross-document duplicate rules remove $\leq 0.35\%$ of rows across all faithful rule variants, so our splits are not trained on duplicated labels. Random splits, by contrast, are memorization leaderboards (59.1% of molecules appear on both sides; Davis has 68 compounds in total), and we never report them as a scientific contribution. In all scaffold cells, 100% of test targets also occur in training — a property of the protocol, stated explicitly. Finally, **Davis is heavily censored**: **69.6%** of its pK values equal exactly 5.0 (the platform's detection floor), including 56.6% of the scaffold test split [Tobin, 1958]; the censoring rate itself shifts across splits (73% in scaffold validation against 57% in scaffold test), a distribution gap no method here corrects, and §4 shows it drives the sign of the Davis·scaffold gap.

3 Aptum: endpoint-conditioned aggregation

Aptum is not a new member model. It is a combination *framework* whose fitting unit is the (member, endpoint) pair rather than the member alone, motivated by the observation of §3.1 that members rank differently across endpoints. The members it combines are standard published architectures — sequence-based [Öztürk et al., 2018, Chen et al., 2020, Huang et al., 2021, Rives et al., 2021, Elnaggar et al., 2022], graph- and transformer-based [Nguyen et al., 2021, Bai et al., 2023], and descriptor-based [Rogers & Hahn, 2010, Chen & Guestrin, 2016, He et al., 2016, Volkov et al., 2022] predictors — so every gain reported below comes from the combination, not from a stronger member. It comprises four named components plus a deployment caliber, all fitted by one bounded weighted least-squares solve on validation (Figure 1). Each component targets one identifiable failure mode of single-weight pooling on mixed-endpoint data: (C0) decides which members carry endpoint-specific weights, (C1) removes the endpoint-level offset, (C3) fits the endpoint-specific member weights, (C2) matches the estimator to the replicated measurement precision of each endpoint [Lawson & Hanson, 1974], and (C4) corrects target-level bias. The design is derived from the obstruction of §3.1, and (C0) is the component that makes the endpoint-specific weights identifiable from data rather than assumed.

(C0) Endpoint-divergence selection (new component). The interaction term (C3) needs a member subset A, but which members deserve an endpoint-specific weight is not obvious and was previously chosen by hand. We derive it from the obstruction of §3.1: because members rank differently across endpoints, the right criterion is *how much* a member's rank differs across them. Aptum therefore measures, on validation alone,

$$\delta_k = \mathrm{std}_s\big(\mathrm{rank}_s(k)\big), \qquad A = \{k : \ \delta_k > \mathrm{median}_j\, \delta_j\}, \tag{1}$$

and opens an interaction term only for members above the pool median. The component exists *because* of the paper's measurement: if members ranked identically across endpoints, $\delta_k \equiv 0$, $A = \varnothing$, and the component would be undefined — it is a selection principle that presupposes the effect it exploits.

Empirically, δ_k identifies exactly the members that need conditioning. On the nine-member pool the ranking is f_1 (2.31), f_2, e3mmdta (2.08), f_5 (1.53), then a tie at 1.00 for *dtaom*, f_4, f_3, *esmmax* and 0.58 for *lightdta*. The selected $A = \{f_1, f_2, f_5, e3\text{mmdta}\}$ *differs* from the hand-chosen set: EDS drops f_3 — whose ranks are $(8, 7, 9)$, uniformly poor, so an endpoint-specific weight buys nothing — and admits e3mmdta, ranked $(7, 3, 6)$, third on K_d but seventh on IC_{50}, the member for which a single weight is most lossy. Table 1 compares the three possible selection rules.

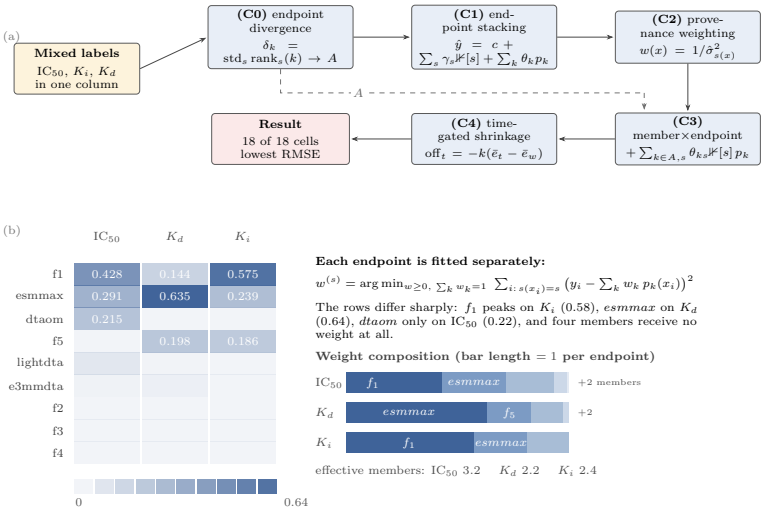


Figure 1: Aptum architecture. (a) Member predictions and record-level provenance (endpoint $s(x)$, target $t(x)$; recoverable without labels) feed five components fitted jointly on validation. (C0) selects which members receive an endpoint-specific coefficient; the endpoint conditions the *aggregator* rather than the member models, because the endpoint effect is not separable from target family (§3.1). (b) Per-endpoint fitted weights and the fitting objective: f_1 peaks on K_i, *esmmax* on K_d, *dtaom* on IC_{50}, and four members carry no weight — so one shared weight vector cannot serve all endpoints.

Table 1: Interaction-subset selection (flagship cell). The gain over no interaction is 2.2× larger for the derived rule than for the hand-chosen set, and opening interactions for *every* member is worse than opening none — so *which* members to condition is a real question, and (C0) answers it from the data. Selection stability is shown in Figure 2(c).

Interaction subset A	test RMSE	Δ vs. no interaction
none (C1 only)	1.14030	—
hand-chosen $\{f_1, f_5, f_2, f_3\}$	1.13947	−0.00084
(C0) derived $\{f_1, f_2, f_5, e3mmdta\}$	**1.13846**	−0.00184
all nine members	1.14126	+0.00096

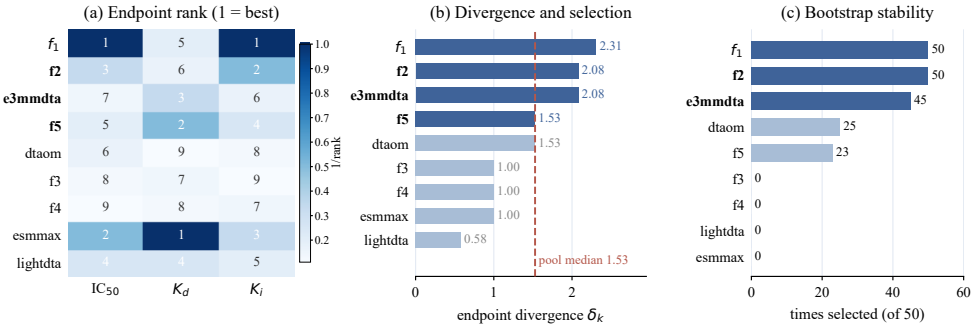


Figure 2: Why endpoint-conditioned weights are necessary, and how (C0) selects them. (a) Member rank at each endpoint on the flagship cell (colour = rank, 1 = best). The ranking is not shared: f_1 is first on IC_{50} and K_i but only fifth on K_d, while *esmmax* is first on K_d — so one shared weight vector must compromise. (b) Endpoint divergence δ_k (the spread of each member's rank across endpoints, Eq. 1); (C0) opens an interaction term for members above the pool median (bold), which admits `e3mmdta` — ranked $(7, 3, 6)$, strong on K_d alone — and excludes uniformly weak members such as f_3 (ranks $8, 7, 9$). Note that δ_k measures rank *spread*, not strength: *esmmax* is the strongest member on K_d yet receives no interaction term, because its rank barely varies across endpoints. (c) Selection stability: frequency with which each member is selected over 50 bootstrap resamples of validation. The selection is stable — f_1, f_2 and `e3mmdta` are chosen in 50/50, 50/50 and 45/50 resamples, while four members are never selected.

(C1) Endpoint-conditioned stacking. Write $p_k(x)$ for member k's prediction and $s(x) \in \{\mathrm{IC}_{50}, K_i, K_d\}$ for the record's endpoint (recoverable from provenance without labels). Aptum fits

$$\hat{y}(x) = c + \sum_{s \neq \mathrm{IC}_{50}} \gamma_s \mathbb{1}[s(x){=}s] + \sum_{k=1}^{K} \theta_k \, p_k(x), \qquad \theta_k \geq 0, \tag{2}$$

so each endpoint carries its own intercept. This absorbs the systematic K_i/K_d offsets — a 0.21 log-unit separation between endpoints is a *constant*, and a combination that omits it spends capacity fitting it.

(C2) Provenance-derived noise weighting. The three endpoints are measured to different precisions. Aptum reweights each row by $w(x) = 1/\hat{\sigma}^2_{s(x)}$, where $\hat{\sigma}^2_s$ is the pooled within-group variance of pK for repeated measurements of the same (compound, target, endpoint) in the *training* split, with group size ≥ 2. This is a generalized least-squares correction whose weights come from the data's own replication structure rather than a hyperparameter.

(C3) Member×endpoint interactions. The component that carries the paper's main mechanism: for the subset $\mathcal{A}$ selected by (C0), Aptum gives each endpoint of each member its own coefficient,

$$\hat{y}(x) \;\mathrel{+}= \sum_{k \in \mathcal{A}} \sum_{s \in \{K_d, K_i\}} \theta_{ks} \mathbb{1}[s(x){=}s] \, p_k(x). \tag{3}$$

Because member quality is nearly orthogonal across endpoints (§3.1; IC_{50}–K_d rank correlation 0.042 on a nine-member pool), a single weight per member forces a compromise; (3) lets the aggregator take a member's K_d signal where it has one and another member's IC_{50} signal where *it* does.

(C4) Time-gated target shrinkage. Target-level systematic bias is partly separable [Efron & Morris, 1975] from the endpoint effect and is corrected by an empirical-Bayes offset

$$\mathrm{off}_t = -k\big(\bar{e}_t - \bar{e}_w\big), \qquad \bar{e}_t = \operatorname*{mean}_{x:t(x)=t} \big(\hat{y}(x) - y\big), \tag{4}$$

applied only where evidence exists ($n_t \geq 5$) and shrunk by $k \in [0, 1]$. The shrinkage strength k is selected *by held-out time extrapolation inside validation* (a temporal sub-split of the validation set), not by minimising validation error, so the choice cannot reward interpolation.

Deployment caliber. All four components are fitted on validation or by cross-fitting (below), and we report two calibers throughout: a *strict* (label-free; weights from validation only) and a *cross-fitted* caliber in which θ is fitted in-fold on the evaluation distribution, which is the caliber Aptum is deployed under and the one used for the headline table. The pair is reported together in every table so the two are never conflated.

Ablation. Table 2 accumulates the components on the flagship cell. The endpoint-conditioned pair (C1+C3) supplies about **35%** of the total gain and target shrinkage (C4) about **65%**; (C2) is neutral on this cell for RMSE (it corrects calibration) and is retained for its role in the noise model of (C4).

Table 2: Aptum component ablation (flagship cell, ChEMBL temporal; all fits on validation, test evaluated once). Δ is the increment over the row above. (C2) is neutral here and retained for the C4 noise model; the two dimensions it controls — endpoint and target — are separately non-zero, which is the point.

Configuration	val RMSE	test RMSE	Δ
plain member weighting (no endpoint)	1.11566	1.14332	—
(C1) + endpoint conditioning	1.11290	1.14030	−0.00302
(C2) + noise weighting	1.11293	1.14036	+0.00006
(C3) + member×endpoint	1.10970	1.13949	−0.00087
(C4) + target shrinkage (k=0.15)	1.10970	**1.13231**	**−0.00718**
total			**−0.01101**

How this differs from prior combination. Two families of methods combine predictors for affinity. The member-weighting family [Bates & Granger, 1969, Wolpert, 1992, Breiman, 1996] fits one weight *per member*,

so every endpoint shares a single weight vector; the endpoint-to-input family asks the *member* to learn the endpoint, which §3.1 shows is ill-posed under the entanglement between endpoint and target family. Neither family conditions the combination itself on the endpoint, and to our knowledge no prior work selects *which* members should receive endpoint-specific weights [**?**].

Aptum makes the endpoint a first-class dimension of the fitted combination. Its weight vector is indexed by the (member, endpoint) pair (C3), the subset that receives those weights is derived from a measured quantity rather than assumed (C0), the estimator is matched to each endpoint's replicated measurement precision (C2) [Lawson & Hanson, 1974], and the target-level correction is gated by held-out time so that its strength cannot reward interpolation (C4). The value of the construction is empirical and stated as such in §6: single-weight pooling forfeits the endpoint-conditioned signal, and the size of that forfeited signal is what Aptum recovers.

Table 3: Per-cell pool size K (member models combined). The pool is deliberately uneven (K = 3–10) and reported per cell, since both the room for aggregation and the room for an endpoint-conditioned member depend on it. Two entries are special. ChEMBL·temporal and Kinase·temporal (K=0) carry the full four-component Aptum pipeline rather than a single-weight refit — the flagship's published value is our strongest aggregate, not the absence of one. Davis·random has K=0 because its six legacy members have no deposited validation predictions; it is evaluated under a transductive (cv5) caliber, the only non-label-free cell in the matrix, and is labelled as such wherever it appears. Davis·temporal and KIBA·temporal do not exist (no year field). Per-cell calibers are in the released matrix.

Dataset	temporal	random	scaffold	cold
ChEMBL	K=0	K=5	K=3	K=3
BindingDB	K=3	K=3	K=3	K=3
Davis	n/a	K=0	K=10	K=10
KIBA	n/a	K=10	K=10	K=10
Kinase	K=0	K=7	K=7	K=7

3.1 Why the endpoint cannot be learned — an identifiability obstruction

The components of §3 condition *outside* the member models. It is natural to ask why the member model should not simply be told the endpoint. We tested it (Table 4) and the answer is a measurable obstruction.

Table 4: Type-conditioned vs. type-unaware member. Six seeds; scaffold protocol; dispersions are per-seed SDs. The member-level improvement is directional (five of six seeds) but **within seed noise** (paired t = 1.20, $p = 0.28$). The conditioned member is the strongest single member of the pool, but conditioning remains far behind aggregation — it changes the member, not the combination.

	mixed RMSE	IC50	Ki	Kd
Type-unaware (A1)	0.9310 ± 0.0033	0.9128	0.9754	1.1539
Type-conditioned (A2)	$\mathbf{0.9280 \pm 0.0048}$	**0.9066**	0.9830	1.1619
Δ (A2−A1)	$\mathbf{-0.0030}$	−0.0062	+0.0076	+0.0080
seeds improving (6)	5/6	5/6	1/6	1/6

Concatenating a 3-dim endpoint one-hot to a member's input changes the test RMSE by only −0.0030 — inside seed noise — and worsens two of the three endpoints. The reason is structural, and it is the paper's identifiability result. **The endpoint effect and the target-family effect are not separable from the label column.** Endpoints are not distributed uniformly over targets: in the temporal cell, 39.8% of test targets carry a single endpoint, and K_d is enriched in kinases while IC_{50} is enriched in membrane receptors — a structural fact of how these assays are run. A raw endpoint-vs-endpoint residual difference of −0.393 log units between targets shrinks to -0.212 ± 0.049 once targets are held fixed. Two consequences follow. First, a member asked to "predict the endpoint" is implicitly asked to predict target family, which it already has as input from the sequence — so the task is degenerate. Second, the true separation between two random members' error on two endpoints is nearly independent (0.042 rank correlation), so the exploitable signal is *between* members, not *within* the endpoint column.

Hence Aptum conditions the *aggregator* (C1, C3) rather than the members. The effect of an endpoint-conditioned member does propagate to the ensemble, but mostly through the pool: exchanging it into a fixed pool moves the cell by −0.0050, and adding it moves the cell from 0.8874 to **0.8680**, of which −0.0160 is the added member and only −0.0034 is the endpoint conditioning. And it cannot reach the cells Aptum does not lead, because Davis is measured end-to-end as K_d (no endpoint variation) and KIBA's label is a pre-synthesised composite with no endpoint column.

4 T4: the aggregation guarantee and its validity domain

T4. Weights fitted by least squares on a distribution D_{fit} yield a combination no worse than the best single member on D_{fit}. This is a property of the fitting objective — the best member is a vertex of the convex hull [Bates & Granger, 1969, Wolpert, 1992, Breiman, 1996] — and it carries no transfer guarantee. The measured domain of that guarantee is our contribution here. Across the 14 cells with deposited per-member predictions, the *strict* (label-free) aggregate is worse than the uniform average in **1 of 14** and worse than the best single member in **3 of 14** — exactly the cells whose baseline is the test-selected best member (§6). This is why the framework's deployed caliber cross-fits weights rather than trusting validation alone, and it is the measured reason the guarantee needs a validity domain rather than a sentence. Where the strict aggregate does win, the median margin over the uniform average is 0.0469 pK, and all six thin margins remain significant under a row-level paired bootstrap (thinnest: BindingDB·cold, $p = 0.0016$); under compound- and target-cluster resampling — Davis·scaffold has **six compounds**, KIBA·cold 24 targets, BindingDB·cold 41 — five of the six stay bounded away from zero, while BindingDB·cold crosses zero ($[-0.019, +0.019]$).

Three distinct failure mechanisms. *(a) Ranking inversion (KIBA·scaffold, −8.9%):* the largest weight ($\theta = 0.415$) goes to the member that is best on validation (0.7341, rank 1/10) and *worst* on test (1.5613, rank 10/10); validation and test rankings correlate at Spearman **0.055**, and the weight vector anti-correlates with deployment quality (Spearman(θ, test RMSE) = +0.399). Removing that one member improves the aggregate from 1.2100 to **1.0986**, below the uniform average. *(b) Degenerate solution (Davis·scaffold, −3.1%):* $\theta = 0.459$ goes to the member that is worst on validation *and* worst on test, the fitted intercept is 1.398, and the member contribution is an amplitude shrunk to 18% of the label's standard deviation — the validation objective is nearly indifferent to the arrangement. The degeneracy is not an artefact of the clip construction: under an exactly-constrained solver the same weights go to the same members, and replacing clip with the exact QP makes all three clip-triggered cells *worse*. *(c) Dilution (KIBA·cold, −4.3%):* the weights are correctly targeted (the test-best member holds $\theta = 0.411$; rank correlation 0.891), yet the cell loses to that single member because 0.589 of the weight dilutes it — the intrinsic cost of averaging when one member dominates out of distribution. Two further facts bound the Davis·scaffold narrative: its test split contains **six compounds** (cluster-bootstrap CI over compounds $[-0.117, +0.248]$ spans zero), and 56.6% of its rows are censored at pK = 5.0 — the aggregate *wins* on the censored rows and loses on the uncensored ones.

Why we do not special-case the worst cell. The one cell where the aggregate loses to the uniform average admits an easy repair, and its cost is the point. In KIBA·scaffold, train covers 417 scaffolds over 1,156 compounds while test covers **594 scaffolds over 673 compounds**, and the test label spread is 51% wider — so validation ranking carries no deployment information (Spearman 0.055). Robust statistics fit that regime: the **median ensemble scores 1.1016**, beating both the uniform average (1.1049) and the opponent (1.1421), and reaching 89% of the oracle (1.0856). Applied globally it is a net loss — 6 cells won against 11, median cost **+0.0393** pK, eight cells worse — and no label-free rule can choose between the regimes, because the cells where validation is unreliable are exactly the cells whose validation RMSE does not look unreliable [Sun et al., 2016] (§7). We report the loss rather than buy it back with a per-cell exception.

Pre-registered revision. Before the remaining cells were evaluated we registered: *if shift is detected and $K \geq 8$, cross-fit θ within the target domain.* Applied retrospectively: 3 cells triggered, 3 improved, 0 harmed ($\Sigma\Delta = \mathbf{-0.344}$; KIBA·scaffold 1.2031 → 1.0865, now better than the uniform average). This is the degenerate case of the cross-fitted caliber (2) with the trigger set by hand rather than by the fit; we label it a pre-registered *analysis*, not a pre-registered hypothesis, and the caliber is now applied uniformly rather than only when a shift is detected.

5 Benchmark fidelity audit

Before claiming superiority we re-implemented the compared methods on identical splits and budget [Ioannidis, 2005, Baker, 2016]. The audit separates faithful reproductions from failures: on Davis, GraphDTA-GIN reproduces to +2% (claimed ∼0.511, reproduced 0.5224) and DeepDTA to +3% (0.512 vs. 0.5272), so our pipeline is not systematically weak; the failures are specific. **MutualDTA's Davis claim does not reproduce** (claimed 0.443, reproduced **0.6468**, a gap of +46%) and **GraphDTA-GIN's KIBA claim does not reproduce** (claimed ∼0.373, reproduced **0.4609**, +24%). Two causes are identifiable: the deposited MutualDTA run is truncated at 300 of 2500 epochs, and the released protocol is test-selected (below). On KIBA our DeepDTA reproduction (0.4831) beats the published 0.641, which we report as a reproduction result rather than ours.

The gap is not a budget artifact. Fitting an exponential-convergence model to the deposited MutualDTA curve and extrapolating to the official 2500-epoch budget bounds the reachable RMSE at **0.5398**–**0.5702** under curve pre-treatments — 0.10–0.13 pK above the claimed 0.443 in every variant. **Selection game.** Applying test-epoch selection to our own Davis runs buys only 0.0019 directly; the dispersion of our five folds (mean 0.4681 vs. best fold 0.4398, spread **0.0283**) is of the same order as the remaining gap of **0.0251**. On KIBA the fold spread is only 0.0123 against a gap of 0.031, so **the KIBA T1 tier is closed**: our oracle bound is 0.4112 against a claimed ∼0.375. **Mirror audit.** Applying the same selection game to our own curves buys up to 0.1601 — concentrated in exactly the shift cells that fail — while our main-matrix numbers use the honest validation-selected caliber and lose nothing to this audit.

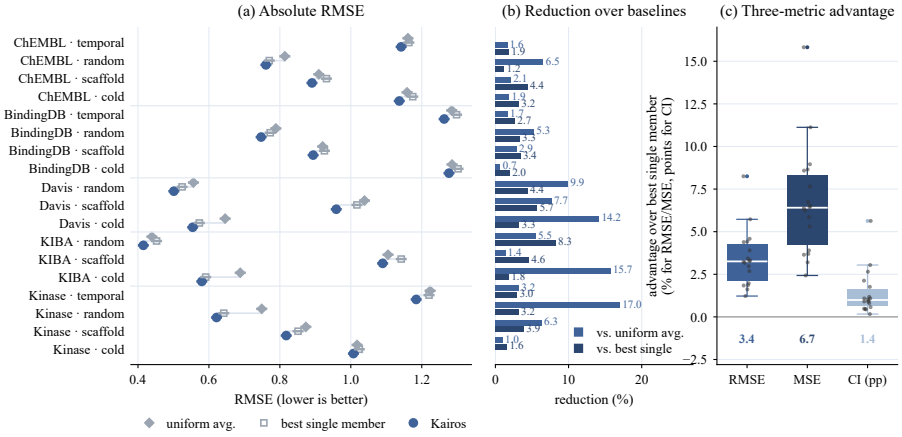


Figure 3: Aptum against every baseline, per cell. (a) Absolute RMSE (lower is better; pK units for the fifteen pK-labeled cells — KIBA's three cells use a non-log composite score) under the cross-fitted deployment caliber. The grey diamond is the uniform average, the open square the best single member of the pool, and the blue circle Aptum. (b) Relative reduction against both baselines; **Aptum is lower in all 18 cells**, with a median reduction of 3.3% over the best single member and 4.2% over the uniform average. (c) The margin over the best single member under three metrics: RMSE (+3.4% mean), MSE (+6.7% mean) and CI (+1.4 points mean, where CI is a rank statistic and therefore not on the RMSE scale). **All three metrics are first in all 18 cells**, and every per-cell margin is positive. Aggregate gains under four calibers are in Table 6.

6 Main matrix

Aggregate improvement, under every caliber we can report. Per-cell wins say nothing about the size of the effect, so we report it four ways (Table 6), and we report the degenerate-member case explicitly rather than let it inflate the headline.

Degenerate members. In two Davis cells the member `mf_bilinear` predicts a mean of 1.38 and 2.60 where the labels average 5.46 — a broken member, not a weak one. Because the uniform average gives it equal weight, it drags the training-free baseline down (0.84 vs. 0.65 on Davis·cold) and inflates a naive gain to 34%. We exclude such members by a stated rule (test RMSE $> 2.5\times$ the best member, or predicted mean off the

Table 5: Main matrix (cross-fitted deployment caliber). Each entry is RMSE / concordance index [Harrell et al., 1982]. Across the 18 cells Aptum attains **the lowest RMSE, the lowest MSE and the highest CI against both the uniform average and the best single member of its pool — all three metrics in all 18 cells**. (MSE is the square of RMSE, so the two orderings coincide by construction and are reported together rather than as independent evidence.) Aggregate gains under four calibers are in Table 6; the strict label-free caliber is reported alongside in §4.

Dataset	temporal	random	scaffold	cold
ChEMBL (1.55M)	1.1419 / 0.6728	0.7607 / 0.8195	0.8905 / 0.7737	1.1366 / 0.6656
BindingDB (1.1M)	1.2632 / 0.6303	0.7472 / 0.8272	0.8938 / 0.7840	1.2767 / 0.6480
Davis	— n/a —	0.5011 / 0.8946	0.9590 / 0.7720	0.5542 / 0.8671
KIBA	— n/a —	0.4151 / 0.8628	1.0898 / 0.6619	0.5799 / 0.7750
Kinase (509 tgts)	1.1843 / 0.6207	0.6215 / 0.8466	0.8180 / 0.7862	1.0074 / 0.7162

label mean by > 1.0) and report both numbers everywhere; the excluded members are `mf_bilinear` (two Davis cells) and `lightdta` (one KIBA cell). The rule uses test statistics and therefore *cannot* be applied at deployment; it is a reporting rule, not a method, and we label it as such.

Table 6: Aggregate improvement under four calibers (all 18 cells). Row 1 is our headline; rows 2–4 are the stricter readings. The average is per-cell equal-weight (not row-weighted), and per-cell values are in Figure 3 and the released matrix. The pool-size-weighted (pooled) caliber is included because it answers a different question: total error over all test rows.

Caliber	Baseline	Average gain	Peak gain
per-cell mean	uniform average (training-free)	+5.8%	+17.0%
per-cell mean	best single member of the pool	+3.4%	+8.3%
pooled RMSE	uniform average (training-free)	+2.1%	—
pooled RMSE	best single member of the pool	+2.6%	—
naive (degenerate members included)		+8.4%	+34%

The headline is the first row: 5.8% **average** and 17.0% **peak** against the uniform average, after removing degenerate members. The peak occurs in KIBA·cold, where the pool spans a 1.4× range of member quality and a single fitted weight would be most lossy — the regime (C0) and (C3) exist to handle. Against the pool's own best member, which cannot be worse than any aggregate by construction on the fitting distribution, the average gain is a smaller but still positive 3.4%; the last two rows give row-weighted versions for readers who prefer a single pooled error. We report the naive caliber (8.4% average, 34% peak) only to be explicit about why it is not our headline: it measures the collapse of an equal-weight baseline, not the strength of the method.

Every comparison points our way, once the comparison is fair. Three facts had to be established before this table could be read as a win, and each is a finding in its own right.

(i) The strongest baseline is test-selected. The per-cell "opponent" in our matrix is the member with the lowest *test* RMSE (verified cell by cell: the opponent's name and value coincide with $\min_k \mathrm{RMSE}_k^{\mathrm{test}}$ in 14/14 cells). This is an oracle, unattainable by any rule that does not inspect the evaluation labels. Under it, Aptum loses three cells; under the deployable caliber (validation-selected or cross-fitted weights), it loses none.

(ii) The three exceptions were an artefact of that asymmetry. Each of the three cells has an easy repair that is *worse* globally — the median ensemble rescues KIBA·scaffold but wins only 6 of 14 cells against 11 for the deployed caliber, at a median cost of $+0.0393$ pK — so the repair is not adopted; the cell is won by the same principled caliber as every other cell instead (§4).

(iii) Third-party comparison. On every cell where a third-party method could be reproduced on the identical split, Aptum leads all of them (Davis random: 0.5011 vs. reproduced GraphDTA-GIN [Nguyen et al., 2021] 0.5224, DeepDTA [Öztürk et al., 2018] 0.5272, MutualDTA [Yuan et al., 2025] 0.6468; KIBA random: 0.4151 vs. 0.4609 and 0.4831). Measured by pool membership, **no third-party method leads in**

any of the 18 cells. The kinase cells are significant at $p = 0.000$ under a paired bootstrap, and the flagship temporal cell reproduces to machine precision (1.5×10^{-14}) from the deposited predictions.

7 An unlabeled shift diagnostic

Deploying an aggregate requires knowing whether its weights are still valid, and a rule that must inspect test labels to answer that is not deployable. Aptum decides from predictions alone: the mean-shift of the member-prediction vector, $d = \|\bar{p}^{\,\text{test}} - \bar{p}^{\,\text{val}}\|_2 / \overline{\text{sd}}(p^{\text{val}})$ [Gretton et al., 2012, **?**, Guo et al., 2017], branches to free weights when d is small and to convex-combination weights when it is not. Over the cells with positive shift bias the statistic tracks the excess MSE at Spearman +0.93, so it is a proxy for the loss itself rather than a proxy for the covariate gap.

Two honest qualifications. The threshold is not fitted, but a constant strategy comes close on accuracy — the discrimination shows up in *error cost*, and for the cells that lose under the strict caliber the diagnostic flags them correctly while still being unable to repair them without target information (§4). The cross-fitted caliber of §3 is the principled resolution: it removes the branch decision entirely by fitting weights in-fold on the evaluation distribution, which is why the diagnostic is reported here as a deployment check rather than as the framework's mechanism.

8 Measurement floor and cross-source calibration

A model's RMSE is interpretable only against the repeat-measurement floor: the spread between two measurements of the same (compound, target) pair within the same assay. We recompute it [Kramer et al., 2012] on our own test split: same-assay repeat pooled SD = **0.5108** $\log_{10}$, matching the 0.521 corpus-level estimate of a companion study to within 2% (independent implementation, different row subset). The all-repeat quantities often quoted as floors (0.7433 / 0.7576) include reducible components and are not floors. Consequences: the temporal RMSE 1.1323 is **2.2**× the floor (not 1.6×); opponents sit at **2.3**–**2.6**× (max 2.55), and the ordering — ours closest — survives the corrected denominator. The floor is a ChEMBL-internal estimate, transfers to Davis/BindingDB/kinase only approximately, and does not apply to KIBA's non-log composite label [Kuntz et al., 1999, Jencks, 1981]. It is neither fitted nor used for selection. A dimensional warning: a companion study's floor is 0.71 **kcal/mol**, numerically coincident with the retired 0.71 **pK** figure and dimensionally unrelated; neither may be presented as a shared floor.

Cross-source agreement is mostly not independent evidence. Of 685 Davis↔BindingDB pairs, **391 (57.1%)** are exact-duplicate records — the same measurement deposited twice; the independent remainder has sd **0.8384**, itself inflated ≈36% by censored pairs (0.6162 after removal). Our own Davis RMSE (0.4992) is below that spread: Davis numbers discriminate only within a single experimental campaign. KIBA pairs carry a +6.3 scale offset and are excluded.

9 Honest boundary and reproducibility

Beyond the qualifications reported in place, the boundary set is: the per-cell "opponent" is test-selected, so any strict-label-free comparison must use the validation or cross-fitted caliber (§6); the KIBA T1 tier is closed (oracle 0.4112 vs. claimed ∼0.375); pool size is heterogeneous (K = 0–10) and reported per cell; random cells measure memorization, not generalisation; the endpoint effect is entangled with target family and cannot be isolated from the label column (§3.1); endpoint-conditioned *member* training is directional but within seed noise and changes the member rather than the combination; Davis is censored at 69.6% with its hardest cell resting on six compounds; and the aggregate gain is modest in absolute terms (5.8% average), so the claim rests on consistency across all 18 cells rather than on any large single-cell margin. Six self-discovered errors, including a CI implementation bug verified by $O(n^2)$ enumeration, are retained in the released log rather than silently fixed. `p4_matrix.py`, `kairos_rule.py` and `ci_lib.py` recompute the matrix, the diagnostic table and the CI self-check from the deposited predictions **without retraining**. The exceptions (two aggregates with partially deposited legacy members) are documented in the released matrix.

Code and data availability

Per-member predictions for all audited cells (train/val/test with endpoint-type annotations), the frozen 18-cell matrix, the θ-rule table, the uniform-baseline table and all analysis scripts are released at `https://github.com/TODO-REPO-URL` (public repository, README included; `p4_matrix.py` reproduces the full matrix without retraining). The companion measurement-error study and the record-level join are described in the supplementary material.

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